

DRB GARMENTS

Technical Architecture, System Design & Wireframe Specification

Platform: Multi-Tenant Nigerian Fashion OS

Target Stack: React / Next.js / Tailwind / Firebase

1. Executive Summary & Design System

DRB GARMENTS is a digital operating system engineered specifically for Nigerian fashion houses, garment production facilities, and custom tailoring enterprises. The system combines localized operational mechanics (handling Naira multi-method payment flows, fabric tracking, tailor queue management, and grace-period facility governance) with a modern multi-tenant SaaS architecture.

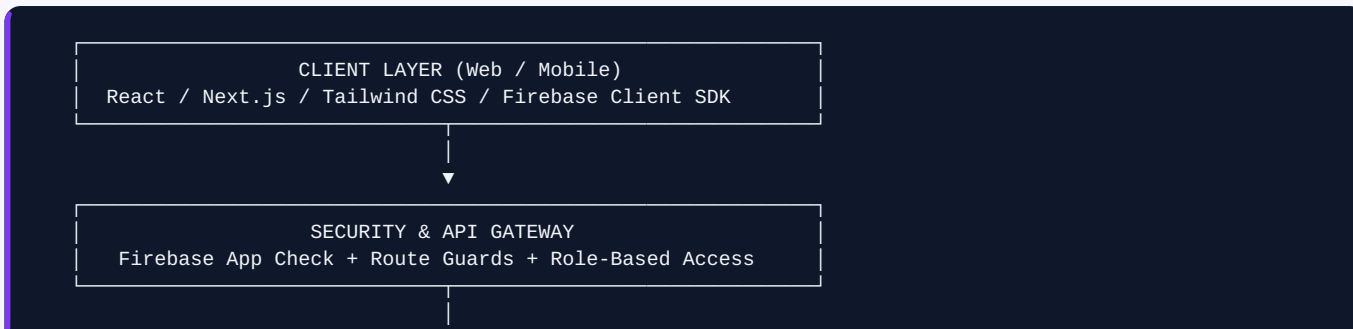
Global Design Tokens: Uses soft lavender/purple backgrounds (#F8F6FF), deep navy typography (#0F172A / #1E1B4B), rounded content cards (16px radius), purple-to-lilac accents, and distinct status indicators for operational visibility.

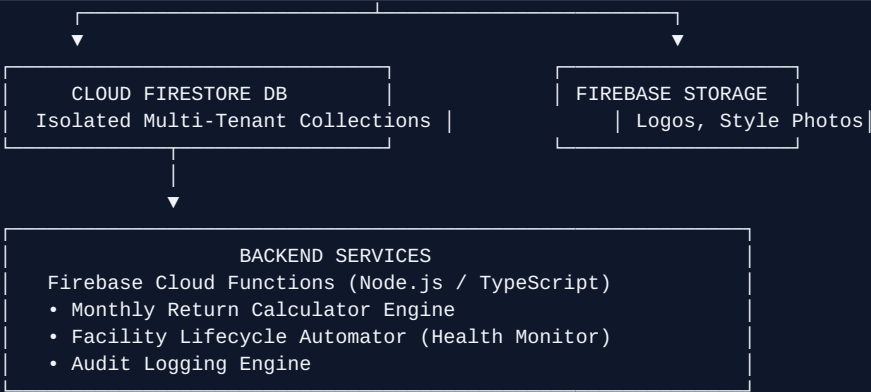
Visual Identity Tokens

Element	Specification	Hex / CSS Value
Primary Background	Soft Lavender & Cool Tints	#F8F6FF / #F1EDFF
Card Surface	Pure White with Soft Shadows	#FFFFFF
Primary Typography	Deep Navy / Dark Slate	#0F172A / #1E1B4B
Brand Accent	Purple to Lilac Gradient	linear-gradient(135deg, #6D28D9, #8B5CF6)
Status Colors	Green (Active) / Amber (Due) / Red (Paused)	#10B981 / #F59E0B / #EF4444
Typography Stack	Modern Sans Serif	Inter, Manrope, system-ui

2. System Architecture & Infrastructure

The application is structured into a client-server architecture powered by Firebase Cloud Services, ensuring real-time synchronized state, role-based security, and strict tenant isolation.





3. Wireframes & Layout Specifications

3.1 Landing Page & Onboarding Hub (`/` & `/onboard`)

DRB GARMENTS	Features	How It Works	Financials	[Sign In]	[Onboard]
Run Your Fashion Business With Precision. From measurements to production, inventory, payments, and profitability – manage your entire fashion house from one intelligent platform.					
[Onboard Your Fashion House]		[Sign In to Workspace]			
FLOATING DASHBOARD PREVIEW CARD					
Active Jobs: 29		Monthly Revenue: ₦4,850,000.00		Staff Active: 14	

3.2 Facility Operating Workspace (`/facility/dashboard`)

BANI COUTURE [Logo]		Search jobs, clients... (User: Owner)				
Powered by DRB Garments						
[X] Dashboard		TODAY'S SUMMARY				
[] Jobs		[Orders Today: 24] [Active Jobs: 18] [Pending Bal: ₦450,000]				
[] Clients						
[] Financials		FINANCIAL PERFORMANCE (SEPTEMBER 2026)				
[] Inventory		Gross Revenue: ₦4,850,000 Purchases: ₦1,250,000 Opex: ₦480,000				
[] Staff		Net Operating Result: ₦3,120,000.00				
[] Settings						
		ACTIVE PRODUCTION QUEUE				
		JOB ID	CLIENT	GARMENT	TAILOR	STATUS
		JOB-2026-081	Chief Adeleke	3-Pce Suit	Emeka O.	[Sewing]
		JOB-2026-082	Amaka Bello	Agbada Set	Yusuf K.	[QC]

3.3 Financial Ledger & Cash Movement (`/facility/financials`)

REVENUE vs PURCHASES vs OPERATING EXPENSES	
SUMMARY CARD:	
Total Money Generated: ₦4,850,000.00 (Customer Deposits + Final Invoices Paid)	

```
| Less Material Purchases: ₦1,250,000.00 (Fabrics, Trims, Accessories, Beading) |
| Less Operating Expenses: ₦ 480,000.00 (Gen Fuel, Electricity, Transport) |
| = Net Operating Result: ₦3,120,000.00 |
+-----+
| RECENT MOVEMENT LEDGER |
| DATE      TYPE      DESCRIPTION      AMOUNT      LINKED REF |
| 11/09/2026 INCOME    Customer Deposit  +₦150,000.00 INV-2026-091 |
| 10/09/2026 PURCHASE  12 Yds Italian Wool  -₦ 85,000.00 JOB-2026-081 |
| 10/09/2026 EXPENSE   Diesel for Workshop Gen -₦ 45,000.00 OPEX-8812 |
+-----+
```

3.4 Lead Admin Command Center (`/admin/dashboard`)

```
+-----+
| LEAD ADMIN COMMAND CENTER (User: Super Admin) |
+-----+
| METRICS: [ Facilities: 148 ] [ Active: 137 ] [ Due Soon: 18 ] [ Paused: 5 ] |
+-----+
| FACILITY CONTROL PROFILE: BANI COUTURE |
| Status: ACTIVE | Subscription: CURRENT |
|
| OPERATIONAL ACTIVITY (Last 30 Days) MONTHLY RETURN (SEPTEMBER 2026)
| • Orders Received: 248 • Transactions: 184
| • Jobs in Production: 29 • Billable Transactions: 161
| • Active Staff: 14 • Platform Charge Rate: ₦50/trans
| • Last Activity: 8 mins ago • Amount Due: ₦8,050.00
| • Status: [ OVERDUE ]
|
+-----+
| ACTIONS: |
| [ Send Payment Reminder ] [ Issue Grace Period ] [ PAUSE FACILITY WORKSPACE ] |
+-----+
```

4. Multi-Tenant Schema & Role-Based Access Control

4.1 Role Authorization Matrix

Feature / Action	Lead Admin	Facility Owner	Facility Manager	Receptionist	Tailor / Sewer
Global Control Desk	FULL	None	None	None	None
Pause / Reinstate Facility	FULL	None	None	None	None
View Revenue & Profit	Read	FULL	FULL	None	None
Purchases & Expenses	Read	FULL	FULL	None	None
Manage Staff Roles	None	FULL	Read	None	None
Jobs & Measurements	None	FULL	FULL	FULL	None
Update Work Queue	None	FULL	FULL	Read	ASSIGNED ONLY

4.2 Security Rules Specification

```
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {

    function isSignedIn() {
```

```

    return request.auth != null;
  }

  function getUserData() {
    return get(/databases/$(database)/documents/users/$(request.auth.uid)).data;
  }

  function isLeadAdmin() {
    return isSignedIn() && getUserData().role == 'LEAD_ADMIN';
  }

  function belongsToFacility(facilityId) {
    return isSignedIn() && getUserData().facilityId == facilityId;
  }

  function isFacilityActive(facilityId) {
    let facility = get(/databases/$(database)/documents/facilities/$(facilityId)).data;
    return facility.status == 'ACTIVE';
  }

  match /admin_audit_logs/{logId} {
    allow read, write: if isLeadAdmin();
  }

  match /facilities/{facilityId} {
    allow read: if isLeadAdmin() || belongsToFacility(facilityId);
    allow write: if isLeadAdmin();

    match /jobs/{jobId} {
      allow read: if isLeadAdmin() || belongsToFacility(facilityId);
      allow create, update, delete: if belongsToFacility(facilityId) && isFacilityActive(facilityId);
    }

    match /ledger/{ledgerId} {
      allow read: if isLeadAdmin() || (belongsToFacility(facilityId) &&
        getUserData().role in ['FACILITY_OWNER', 'FACILITY_MANAGER']);
      allow write: if belongsToFacility(facilityId) && isFacilityActive(facilityId) &&
        getUserData().role in ['FACILITY_OWNER', 'FACILITY_MANAGER'];
    }
  }
}

```

5. Sequential Implementation Roadmap

Phase 1 — Public Portal, Onboarding & Auth Gateway

- Build landing page with floating UI panels and value sections.
- Implement multi-step onboarding wizard storing requests in Firestore (UNDER_REVIEW).
- Configure Firebase Authentication (Google OAuth + Email/Password).

Phase 2 — Isolated Facility Workspace & Financial Engine

- Implement facilityId tenant isolation across all collections.
- Build garment production pipeline (Reception → Cutting → Sewing → QC → Pickup).
- Construct non-destructive measurement version control repository.
- Build Financial Ledger (Money Generated vs. Material Purchases vs. Opex).

Phase 3 — Platform Control Desk & Monthly Returns Engine

- Build Lead Admin Monitoring Hub for platform-wide activity tracking.
- Construct Monthly Return calculation engine and remittance tracking.
- Implement non-destructive facility pause state (read-only preservation).
- Hook up automated audit logging for all administrative actions.